U-W-00149a(DSA-DM)

9 September 1966
Interim Revision of
Fed. Spec. U-W-149
23 November 1948

INTERIM FEDERAL SPECIFICATION

WAX, DENTAL, STICKY

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification U-W-149.

1. SCOPE

1.1 Scope. This specification covers sticky wax suitable for holding various molding materials in dental prosthetic procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6520

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The sticky wax shall be a mixture of waxes and resins or other ingredients combined in proportions that will result in a material possessing the physical properties hereinafter specified.

3.2 Description. The sticky wax shall be suitable for use in prosthetic dentistry. The wax shall be sticky when melted, and shall have good adhesive characteristics. The wax shall be brittle at room temperature and fracture cleanly when broken. The wax shall be furnished in smooth sticks not less

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than 3-3/4 nor more than 4-1/8 inches in length. The cross section shall be round or hexagonal and shall be 1/4 ± 1/16 inches in diameter, or 1/4 ± 1/16 inches across the flats.

3.3 Color. The color shall be dark red or dark violet so as to be readily distinguishable from the white, tan, gray or pink surfaces of artificial dental stones and plaster to which the wax is applied.

3.4 Physical properties.

3.4.1 Flow. The flow of the wax shall be not more than 5.0 percent at a temperature of 30 ± 0.1 degrees centigrade (°C.) and not less than 88.0 percent at a temperature of 43 ± 0.1°C. when tested as specified in 4.4.1.

3.4.2 Ash. The ash content shall not exceed 0.2 percent when tested as specified in 4.4.2.

3.4.3 Shrinking. A wax specimen shall not shrink more than 0.5 percent between the temperatures of 43°C. and 28°C. when tested as specified in 4.4.3.

3.5 Weight. Sticky wax shall be furnished in unit boxes containing a minimum of two ounces of material.

3.6 Lot or batch number. Each unit package shall be marked with a lot or batch number and date of manufacture.

3.7 Stability. The wax shall conform to the requirements of the specification for not less than one year from date of delivery to the Government, when stored in a cool, dry place (50 to 80 degrees Fahrenheit (°F.)).

3.8 Delivery schedule. Not more than two months shall have elapsed from date of manufacture of the wax to date of delivery to the Government.

3.9 Workmanship. Workmanship shall be first class throughout. The sticky wax shall be free from defects which detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted as indicated in table I. The unit of product for inspection purposes shall be one package of sticky wax.

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Table I. Sampling For Tests.

1	2	3	4	5	6
Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations per lot or batch number
Flow	3.4.1	4.4.1	1	3	3
Ash test	3.4.2	4.4.2	1	3	3
Shrinking	3.4.3	4.4.3	1	3	3
Weight	3.5	S.M.I	1	3	3

4.2.3 Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The sticky wax shall be examined to determine compliance with specification requirements.

4.4 Tests.

4.4.1 Flow.

4.4.1.1 Preparation of specimens. A quantity of wax shall be broken into pieces and placed in a metal pouring pan. The pan shall be placed on a surface which is 130 millimeters (mm.)(5.1 inches) below a 250-watt infrared lamp. The wax, while being stirred, shall be allowed to reach a temperature of 75 ± 5°C. (167 ± 9°F.) and be maintained at this temperature until the sample is melted throughout. A thermometer shall be used to measure the temperature. The melted wax shall then be poured into a mold that has been lubricated with a silicone grease whose melting point is higher than 75 ± 5°C. (167 ± 9°F.). The mold shall consist of a stainless steel plate 6.0 mm. (0.236 inches) thick, having flats, parallel top and bottom surfaces, and containing four holes 10.0 mm. (0.394 inches) in diameter. The axes of the holes shall be perpendicular to the surfaces of the plate. The sides of the holes shall be finished smooth. The mold shall be preheated to a temperature of 55 ± 5°C. (131 ± 9°F.) and placed on a smooth glass slab 152 mm. long, 76 mm. wide and 19 mm. thick (6 X 3 X 0.75 inches) preheated to the same temperature. As the wax freezes and a shrinkage void appears, liquid wax

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shall be added. When the wax has lost its "mirror-like" surface, a smooth, flat tin-foil or aluminum-foil covered glass plate, preheated to 55 ± 5°C. (131 ± 9°F.), shall be placed on the top of the mold. A load of 9,000 grams (20 pounds) shall be applied to the top of the foil-covered glass plate for 30 minutes. The weight and the glass plate shall then be removed and the excess wax trimmed away. This may be accomplished by drawing the specimen flush with the surface. The mold shall be removed from the glass slab by gently-tapping the side of the mold. The specimens of wax shall be removed from the mold by chilling in water at 10°C. (50°F.), and shall then be stored at 23.0 ± 2.0°C. (73.4 ± 3.6°F.) for 24 hours before testing.

4.4.1.2 Method of test. The initial length of the specimen shall be determined at 23.0 ± 2.0°C. (73.4 ± 3.6°F.) using a metric micrometer caliper. Four measurements shall be made around the circumference and one measurement shall be made in the center of the specimen. The measurements shall be averaged and recorded to the nearest 0.005 mm. The specimen and flow testing instrument shall be placed in a water bath and held at the testing temperature for 20 minutes prior to testing. The flow testing instrument shall consist of the following: A, a weight; B, a shaft having low thermal conductivity; and C, a brass platen. The total weight, in air, of these three components shall be 2,000 grams (4.41 pounds). The weight, A, shall be separated a minimum distance of 76 mm. (3 inches) from the brass platen by the non-conducting shaft. This shaft shall be of hard rubber or a similarly poor thermal conductor to reduce loss of heat from the specimen. The diameter of the brass platen shall not be less than 51 mm. (2 inches). The thickness shall not exceed 6.35 mm. (0.25 inches). The temperature of the bath shall be controlled to within ± 0.1°C (± 0.2°F) of the required temperature, a calibrated thermometer being used for determining the temperature). Agitation shall be provided in the water bath by means of a mechanical stirrer. A thin sheet of waterproof cellophane shall be placed between the instrument and each end of the specimen. The bottom of the specimen shall be 51 mm. (2 inches) below the surface of the water in the bath. A constant axial load of 2,000 grams (4.41 pounds) shall then be applied to the specimen for 10 minutes.

4.4.2 Ash test. A one-gram specimen of wax shall be ignited in a porcelain crucible at 930 to 950°F. and the residue calculated. The crucible shall be previously ignited to constant weight.

4.4.3 Shrinking test. A specimen 20 centimeters in length shall be conditioned for 20 minutes in water at a temperature of 43°C. The water shall be cooled gradually to 28°C, and the change in length determined by observation with a comparator microscope.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Two (2) ounces of wax, as specified, constitutes one unit of issue, one box.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Unit Package. Each unit shall be packaged in a carton of appropriate size, constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Sticks shall be packaged in two (2) or more layers. Layers shall be separated from each other and from carton walls with suitable material to prevent binding or blocking. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2 Intermediate Package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner will not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Marking shall include the legend: "STORE IN A COOL PLACE (50° - 80° F.)"

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend: "STORE IN A COOL PLACE (50° - 80° F.)"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
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Review activities:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION U-W-00149a(DSA-DM)

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426

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